

Who Bears the Congestion Price?

Executive summary — the burden of NYC's CBD congestion fee and its relationship with taxi and Uber/Lyft trip volume.

Background

On January 5, 2025, New York City implemented the [first congestion-pricing program in the United States](#), charging vehicles that enter a defined zone — the Manhattan Central Business District south of 60th Street (the "Congestion Relief Zone," or CRZ). The fee depends on the vehicle: during peak hours, a private passenger vehicle with E-ZPass pays \$9, generally once per day, each covered taxi ride pays \$0.75, and each covered Uber/Lyft ride pays \$1.50. Manhattan traffic was [reported to be lighter afterward](#). Given the fee structure, private cars are the policy's main target, while taxis and Uber/Lyft face smaller per-trip charges.

Research questions

We instead study the much smaller per-trip charge on taxi and Uber/Lyft, and ask two linked questions:

- Who bears it? Even though the fee is small and mostly flat, is it felt evenly across riders?
- Did it reduce trips? Did taxi/Uber-Lyft volume fall after the fee, and fall more where a zone was more exposed to the CRZ?

Data

We use public [NYC TLC trip records](#) — roughly 37M Yellow Taxi and 202M Uber/Lyft trips across the study windows. Each record contains trip geography (pickup/dropoff zone and distance) and service-specific financial fields. Yellow reports metered totals and tips; HVFHV passenger cost is reconstructed from public TLC components and may not equal the rider's final app payment. We compare February–June 2024 (before) against the same months of 2025 (after) to reduce seasonal mismatch; January 2025 is excluded as a transition month.

The unit of analysis is zone × direction — each TLC zone is scored separately on its pickup side and its dropoff side, because a zone can behave very differently depending on whether trips start or end there. Yellow and Uber/Lyft are analyzed separately in the primary within-service analyses: their cost fields are constructed differently, and Yellow has a Flex-fare regime shift that affects its raw volume. Model 3 then compares aligned Yellow and Uber/Lyft panels within the same zones.

What we measure

- Burden — the Zone Disruption Score, DS_z . For each charged trip, the CBD fee as a share of

what the trip would otherwise have cost; aggregated to zone × direction. A higher DS_z means the fee is a bigger slice of the fare. It measures the relative fee burden.

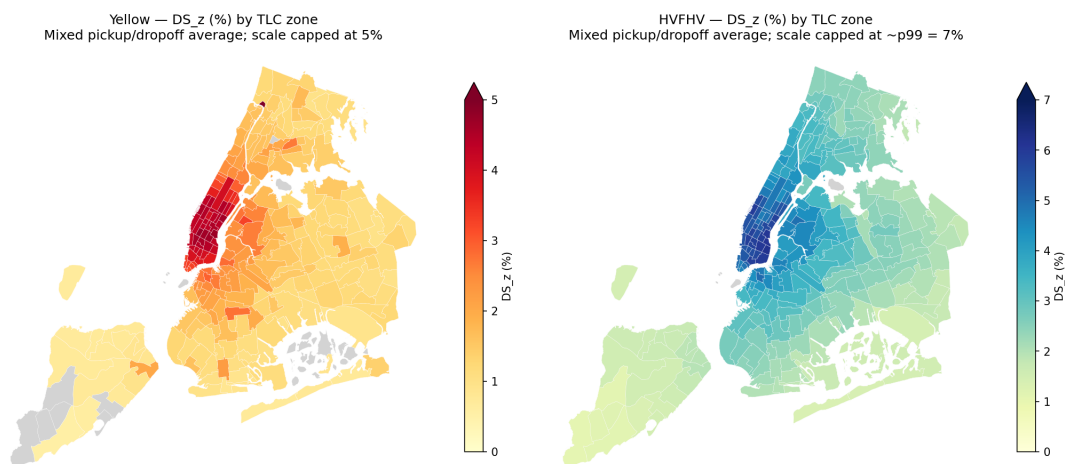
- Outcome — trip-volume change, 2024 → 2025, per zone-side.

Finding 1 — Burden: strong, stable, and uneven

Because the fee is mostly fixed within each service, it accounts for a larger share of the fare on short, low-cost trips than on longer trips. The highest-burden zone-sides are concentrated in the dense Manhattan core for both services (top DS_z $\approx$ 4.6–4.8% for Yellow, $\approx$ 6% for Uber/Lyft); airport trips carry far less ($\approx$ 1–2% vs $\approx$ 4–5% for non-airport trips).

Crucially, the ranking is stable across denominator-floor choices (\$0.50–\$5): Spearman $\geq$ 0.996 for Yellow and 1.00 for Uber/Lyft, with 100% top-10/20 overlap. So DS_z is a reusable, reproducible burden metric that gives a dependable ranking, not an artifact of one reasonable floor choice.

Fee burden (DS_z) by TLC zone — Yellow Taxi (left) and Uber/Lyft (right, labeled "HVFHV"). Both concentrate in the dense Manhattan core. Each service is shown on its own scale — DS_z is comparable within a service, not directly across them.



Fee burden by TLC zone: Yellow Taxi (left) and Uber/Lyft/HVFHV (right). Each service uses its own scale.

Finding 2 — Volume: negative signals, but not a clear causal effect

For the volume question we build a modeling ladder using three regression designs, each with its own controls and assumptions. Each design tightens part of the previous comparison, and Models 2 and 3 are also checked against a no-fee placebo window:

- Model 1 (higher-burden zones vs. lower-burden zones, cross-zone association). Higher-burden zones grew less — strong for Uber/Lyft

($r \approx -0.61$), weak for Yellow ($r \approx -0.17$) — but high burden $\approx$ dense core $\approx$ short trips, so it is confounded with geography.

- Model 2 (higher-exposure zones vs. lower-exposure zones, exposure difference-in-differences). Main equal-weighted estimates are

approximately -11% for both services, but both also show negative estimates in the no-fee 2023→2024 placebo window. Yellow is additionally sensitive to volume weighting and the within-Manhattan comparison.

- Model 3 (Uber/Lyft vs. Yellow Taxi, cross-vehicle DiD, same zones). Uber/Lyft fell $\approx 5.9\%$ more than Yellow inside the CRZ — but the no-fee placebo is large and opposite-signed, and Uber (-11.4%) and Lyft ($+10.3\%$) move in opposite directions despite facing the same \$1.50 fee.

Every design finds a negative main signal, but the diagnostics limit how each estimate can be read. We therefore report the volume evidence as an association, not a causal effect. This is inference, not prediction: the bottleneck is identification — whether the comparison group represents what would have happened without the fee — not model flexibility, so more complex models would not resolve it.

What this adds

- A sharper question — we isolate the taxi/Uber-Lyft layer (who pays the small per-trip charge), distinct from the private-car traffic story.
- A reusable, stable metric — DS_z, reproducible from public TLC data, with a ranking that is stable across the tested denominator floors.
- An honest boundary — placebo tests show that similar exposure-related patterns were already present before the fee, so we mark clearly where public trip data can and cannot support a causal claim.

Main limitations

- TLC data show only completed trips — not mode-switching, cancellations, or trips that were not taken.
- High-burden zones are not random — they are the dense Manhattan core, tied to short trips and low base cost.
- The volume designs are weakened by no-fee placebo checks.
- Yellow's Flex Fare regime shift affects raw Yellow volume; HVFHV cost is reconstructed from public fields and may miss app-side pricing.

Future directions

Three extensions, each targeting a limitation above:

- Sharpen the comparison — longer pre-policy panels, more placebo windows, and boundary or matched-zone checks, for a more credible counterfactual.
- Look beyond TLC trips — transit, bike, vehicle-entry, and traffic-speed data, to distinguish mode substitution from trips that were not taken.
- Separate fee effects from platform dynamics — provider-specific (Uber vs Lyft) baselines and rider-neighborhood context, to better distinguish fee-related changes from Uber/Lyft market shifts.

Each would improve interpretation, but each needs its own design.